LITHO Mainnet 9005 Operations Runbook

Last reviewed: 2026-07-29

Use this runbook for `lithosphere_9005-1` only. Older repository runbooks may refer to decommissioned AWS infrastructure, another chain ID, TMKMS, or a different node home and must not be applied to this mainnet.

1. Safety rules

1. The validator consensus key and `priv_validator_state.json` are a single-writer pair. Never start another signer with the same key.
2. Never replace, reset, or roll back validator signing state to make a node start. Stop and escalate if the correct state cannot be proven.
3. Verify both chain IDs before and after every change: Cosmos `lithosphere_9005-1`, EVM `9005` (`0x232d`).
4. Change one sentry at a time and keep the other sentry healthy.
5. Do not regenerate or edit genesis. The only approved SHA-256 is `13e4875b4a9dddc63bdfbd4968c7265f9bbc49218b59c5b49231a56fa313046f`.
6. Do not replace the approved mainnet binary. Its SHA-256 is `0546677a9cf3a7f458797b65181a46f21c89185933e832d89ce728a144fd258c`.
7. Keep Makalu and Kamet services, homes, ports, and WireGuard interfaces out of mainnet changes.

2. Fast public health check

```
# Cosmos identity, height, time, and catch-up state
curl -fsS https://rpc-mainnet.litho.ai/status | jq '{
  network: .result.node_info.network,
  height: .result.sync_info.latest_block_height,
  block_time: .result.sync_info.latest_block_time,
  catching_up: .result.sync_info.catching_up
}'

# EVM identity; expected result is 0x232d
curl -fsS https://rpc-mainnet.litho.ai \
  -H 'content-type: application/json' \
  --data '{"jsonrpc":"2.0","id":1,"method":"eth_chainId","params":[]}'

# REST identity
curl -fsS \
  https://api-mainnet.litho.ai/cosmos/base/tendermint/v1beta1/node_info \
  | jq '.default_node_info.network'

# Immutable genesis check
curl -fsS https://rpc-mainnet.litho.ai/genesis.json -o /tmp/litho-genesis.json
printf '%s  %s\n' \
  '13e4875b4a9dddc63bdfbd4968c7265f9bbc49218b59c5b49231a56fa313046f' \
  /tmp/litho-genesis.json | sha256sum --check
```

Run the status request twice, five to ten seconds apart. Healthy means the height advances, `catching_up` is `false`, the latest block time is current, and both chain IDs match. A valid HTTP response at a static height is not a healthy chain.

If `grpcurl` is available:

```
grpcurl grpc-mainnet.litho.ai:9090 \
  cosmos.base.tendermint.v1beta1.Service/GetNodeInfo
```

3. Privileged full verification

From an authorized operator workstation with a client-managed SSH key:

```
python scripts\verify_litho_mainnet_9005_live.py `
  --ssh-key "C:\secure-path\litho-validator"
```

The read-only verifier requires strict known-host checking and validates all three services, both chain IDs, peer counts, synchronization, fixed supply, the one bonded validator, seven genesis balances, and the height-1 hash. Retain its timestamped output with the change or incident record. Do not weaken its SSH options to bypass a host-key mismatch; investigate the mismatch.

4. Service map and local checks

Role	SSH host	Service	Local Comet RPC	Local EVM RPC
Validator	`194.5.157.233`	`lithod-mainnet-9005-val`	`127.0.0.1:26657`	`127.0.0.1:8545`
Sentry 1	`31.97.39.146`	`lithod-mainnet-9005-sentry`	`127.0.0.1:27057`	`127.0.0.1:8945`
Sentry 2	`72.60.177.106`	`lithod-mainnet-9005-sentry`	`127.0.0.1:27057`	`127.0.0.1:8945`

On an affected host:

```
sudo systemctl is-active SERVICE
sudo systemctl is-enabled SERVICE
sudo systemctl show SERVICE -p ActiveEnterTimestamp -p NRestarts
sudo journalctl -u SERVICE --since '-30 min' --no-pager
sudo wg show wg-mainnet
curl -fsS http://127.0.0.1:COMET_PORT/status | jq '.result.sync_info'
curl -fsS http://127.0.0.1:COMET_PORT/net_info | jq '.result.n_peers'
sudo sha256sum /usr/local/bin/lithod-mainnet-9005
sudo sha256sum MAINNET_HOME/config/genesis.json
```

Expected peer baseline at handoff: validator `2`; each sentry at least `1`. Check for recent `wg-mainnet` handshakes as well as peer counts.

5. Incident triage

No new blocks for more than two minutes

Treat this as a production incident because one validator currently controls block production.

1. Confirm the halt from two independent endpoints or from both sentries.
2. Freeze planned changes and record the last height and block time.
3. Check validator service state, restarts, disk, memory, time sync, journal, WireGuard handshakes, and peer count.

4. Prove that no other process or host is using the validator consensus key.
5. Preserve the validator journal, config, and current signing state before a change. Never publish or copy key contents into an incident ticket.
6. If the service is stopped and a single writer is proven, start the existing service once; do not launch a replacement signer.
7. Require consecutive block advancement on the validator and both sentries, `catching_up=false`, and a passing full verifier.
8. Open an incident record with impact, timestamps, cause, exact changes, and verification evidence.

The 2026-07-28 restart loop was caused by `double_sign_check_height = 10` and was fixed durably at `0`. Do not rerun the incident recovery playbook as a generic restart procedure: it contains incident-specific height and backup assertions. See the incident record.

One sentry is unhealthy

1. Verify the other sentry is healthy and advancing.
2. Remove the unhealthy sentry from public traffic if necessary.
3. Capture its journal, disk/memory state, peer state, and WireGuard state.
4. Restart only the affected sentry service.
5. Require the correct chain IDs, advancing height, fresh peers, and `catching_up=false` before returning traffic.
6. Do not restart the validator as part of sentry recovery unless it has an independently diagnosed fault.

Both sentries or public endpoints are unhealthy

Check DNS/TLS/Nginx separately from node health. If local node RPCs advance, restore the proxy path without touching consensus. If local node RPCs also fail, handle each sentry one at a time. The validator must remain private.

Suspected consensus-key compromise or duplicate signer

Stop the known validator service, isolate the host, preserve evidence, and invoke the client security and custody owners immediately. Do not destroy the suspected key, rotate it, restore state, or start another signer until a chain-level recovery plan is approved. Key rotation is not merely a file replacement because the on-chain validator identity must also be addressed.

6. Backup and restore controls

Two encrypted offline copies of the launch-time identity package were client-confirmed, but that height-0 package does not contain current signing state. Recurring live signing-state backup is an open Critical gate.

Before operational acceptance, the client must define and test a process that:

- snapshots the current `priv_validator_state.json` consistently;
- encrypts it before leaving the validator and never exposes plaintext in logs, chat, tickets, or version control;
- records a checksum, capture time, chain ID, and signed height;
- stores at least two separately controlled offline copies;
- names primary and backup custodians and retention; and
- performs a restore drill on an isolated host without starting `lithod`.

For a real restore, keep the original signer stopped and prove its signing state is the authoritative latest state before copying the consensus key and state as one controlled unit. If recency or single-writer status is uncertain, do not start a validator.

7. TLS and endpoint operations

At handoff, Let's Encrypt certificates expire on 2026-10-26. Certbot timers are enabled and renewal simulations passed. Operators must still alert on certificate expiry and periodically confirm successful renewal and Nginx reload on both sentries.

The public proxy rate limits RPC to `25` requests/second/IP and REST to `30` requests/second/IP, with bounded bursts. Raw node ports are temporary integration paths, not preferred production interfaces; close them after dependency confirmation while retaining required P2P traffic.

8. Change and maintenance procedure

For every change:

1. Record scope, owner, rollback, and approval.
2. Capture public health and run the full verifier.
3. Confirm current signing-state backup and single-writer ownership for any validator-impacting change.
4. Change sentries one at a time. Schedule validator work in an explicit maintenance window; unattended package restarts caused the launch-day halt.
5. Recheck certificate, chain IDs, height advancement, peers, supply, bonded validator, and height-1 hash.
6. Retain command output and close the change only after an observation window.

9. Escalation record

Complete these fields in the client-controlled copy; do not put personal phone numbers or secret-manager recovery material in the public repository.

Role	Named owner	Primary channel	Backup owner
Incident commander	KaJ Labs	`TBD`	KaJ Labs
Validator operator	KaJ Labs	`TBD`	KaJ Labs
Consensus-key custodian	KaJ Labs	`TBD`	KaJ Labs
DNS/Cloudflare owner	KaJ Labs	`TBD`	KaJ Labs
Monitoring/on-call owner	KaJ Labs	`TBD`	KaJ Labs
Explorer release owner	KaJ Labs	`TBD`	KaJ Labs

KaJ Labs is the accountable organization in this repository copy. Record the individual primary and backup contacts in the client-controlled escalation register. The full control boundary is documented in OWNERSHIP_AND_ADMIN_CONTROL_HANDOVER.pdf.